

Top 50 AI/ML Interview Questions

Machine Learning Fundamentals

1. What is Machine Learning? How is it different from AI and Deep Learning?
2. Explain the complete Machine Learning workflow.
3. Explain Supervised, Unsupervised and Reinforcement Learning with examples.
4. What is overfitting and underfitting?
5. What is bias and variance? Explain the tradeoff.
6. Explain train-validation-test split.
7. What is Cross Validation? K-Fold vs Stratified K-Fold.
8. What are features and labels?
9. Classification vs Regression.
10. What assumptions should you check before training a model?

Data Preprocessing

11. How do you handle missing values?
12. How do you detect and treat outliers?
13. Standardization vs Normalization.
14. Why is feature engineering important?
15. How do you encode categorical variables?
16. What is data leakage?
17. How do you handle imbalanced datasets?
18. Explain feature selection techniques.

Algorithms

19. Explain Linear Regression.
20. Linear vs Logistic Regression.
21. Explain Decision Trees (Entropy/Gini).
22. What is Random Forest?
23. Explain XGBoost.
24. Explain Support Vector Machine.
25. Explain K-Nearest Neighbors.
26. Explain Naive Bayes.
27. Explain K-Means clustering.
28. Bagging vs Boosting.
29. Explain PCA.
30. Which algorithm would you choose for different scenarios?

Model Evaluation

- 31. Explain the Confusion Matrix.
- 32. Precision vs Recall vs Accuracy vs F1 Score.
- 33. When is Accuracy a bad metric?
- 34. Explain ROC Curve and AUC.
- 35. RMSE vs MAE vs MSE.
- 36. Explain Hyperparameter Tuning.

Deep Learning

- 37. Explain Artificial Neural Networks.
- 38. Explain Backpropagation.
- 39. CNN vs RNN vs Transformer.
- 40. Why do CNNs work well for images?
- 41. What is the Vanishing Gradient Problem?
- 42. Explain Dropout.

NLP & LLMs

- 43. Explain Transformer Architecture.
- 44. What is Self-Attention?
- 45. What is RAG? Explain the pipeline.
- 46. LLM vs Fine-Tuning vs Prompt Engineering vs RAG.

Deployment & MLOps

- 47. How do you deploy a Machine Learning model?
- 48. What is Model Drift? Data Drift vs Concept Drift.
- 49. What AI/ML tools have you used?
- 50. Tell me about one AI/ML project you built.

Bonus Advanced Questions

- Why does Random Forest reduce overfitting?
- Explain Gradient Descent mathematically.
- Batch vs Mini-Batch vs SGD.
- L1 vs L2 Regularization.
- Ridge, Lasso, ElasticNet.
- Why do Transformers outperform RNNs?
- Explain Multi-Head Attention.
- What is Beam Search?
- Explain Transfer Learning.
- What is Explainable AI (XAI)?
- How would you monitor an ML model in production?

How would you reduce inference latency?

Design a recommendation system.